

Problem

First fix

Better fix

In practice

Gradient descent
batch, stochastic,
mini-batch
Note ML-056–059

Weak spots of gradient descent

one learning rate,
saddle points
Note DL-032

idea 1: build up speed

Momentum

EWMA of gradients
Note DL-034

NAG

look ahead, then correct
Note DL-035

Exponentially
weighted
moving average
Note DL-033

AdaGrad

own learning rate
per parameter
Note DL-036

RMSPProp

EWMA of
squared gradients
Note DL-037

Adam

momentum +
RMSPProp
Note DL-038

Keras Tuner

tries the
choices for us
Note DL-039

A better
neural network
see the training map
Note DL-021

idea 2: adapt the learning rate

→ fixes

→ used in

→ leads to

problem

from another map